MECHANICAL ENGINEERING
UNIVERSITY OF MINNESOTA

Step Placement Swing Control for Powered Knee-Ankle Prostheses

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Summary

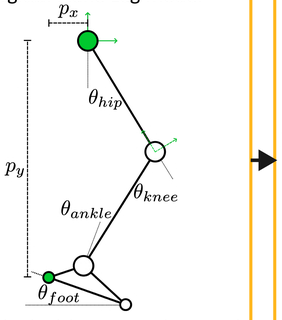
Main Contribution: We designed a novel prosthesis swing-phase controller that encodes human-like step placement behavior and developed a human-prosthesis simulation to validate it against realistic human inputs.

Significance: Step placement is a key component in regulating stability in human walking. This controller coordinates knee-ankle prosthesis joints to achieve human-like stability patterns.

Approach: We developed a data-driven approach to predict desired human step placement in response to the user's hip motion and designed a task-space model-based control method to realize the desired step length. We use human gait data and skeletal models to provide human inputs to a forward-dynamics prosthesis simulation.

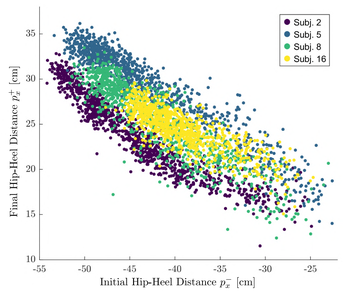
Step Placement Control Method

Sagittal Plane Leg Model



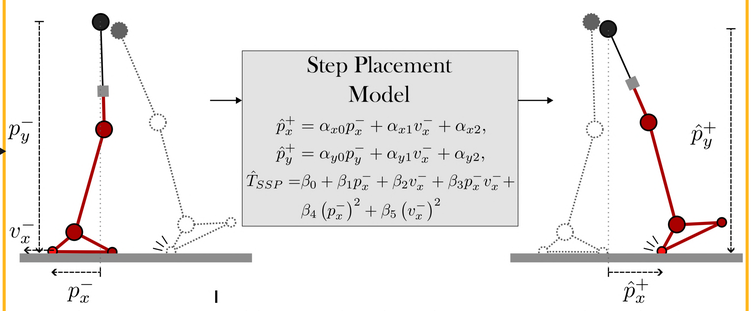
Human hip-heel distances are determined from motion-capture data.

Gait Data Extracted From Recordings



Step placement predictive models are generated for each subject.

Subject-Specific Step Placement Prediction Models

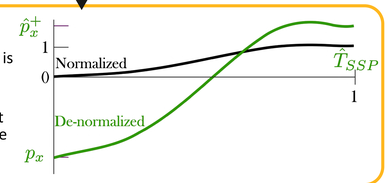


At heel-strike, a predictive model estimates desired step placement position and timing based on relative hip-heel positions and velocities.

Swing-Phase Reference Trajectories

A normalized hip-heel trajectory is generated with subject-specific data.

Trajectories are de-normalized at the start of swing phase to define desired prosthesis hip-heel motion.



Model-Based Task-Space Prosthesis Controller

A task-space feedback linearizing controller is used to track the desired trajectories on the prosthesis.

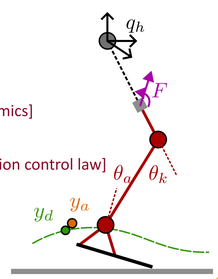
$y = y_a - y_d$ [Output definition]

$$\ddot{y}(q) = \underbrace{\frac{\partial}{\partial q}\left(\frac{\partial y}{\partial q}\dot{q}\right)\dot{q} + \frac{\partial y}{\partial q}\left(D(q)^{-1}(-H(q) + J^T F)\right)}_{L_f^2 y(q,\dot{q})} + \underbrace{\frac{\partial y}{\partial q} D(q)^{-1} B\, u}_{L_g L_f y(q,\dot{q})}.$$ [Prosthesis task-space output dynamics]

$u = L_g L_f y(q,\dot{q})^{-1}(-L_f^2 y(q,\dot{q}) + \mu + \ddot{y}_d)$ [Feedback linearization control law]

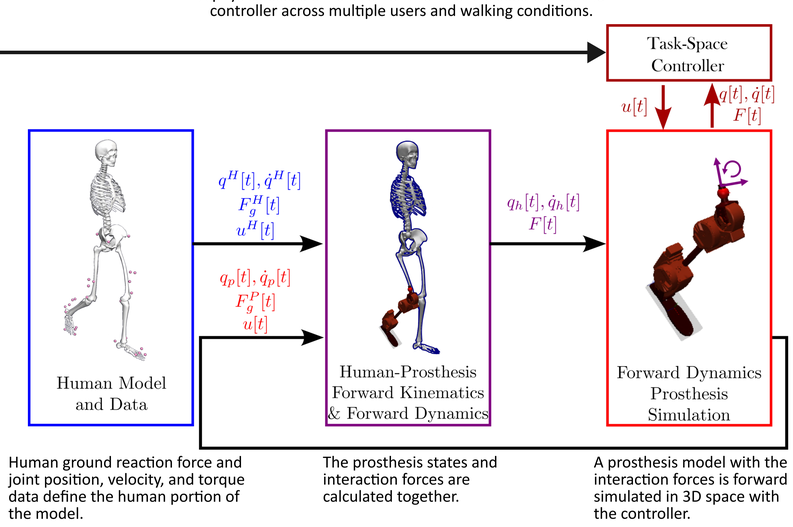
$\mu = -K_P(y_a - y_d) - K_V(\dot{y}_a - \dot{y}_d)$ [PD feedback term]

$L_g L_f y^{-1} = W_q J_W^T (J_W J_W^T + \lambda I)^{-1} W_x,$
$J_W = W_x L_g L_f y W_q$ [Weighted Pseudo-Inverse]



Human-Prosthesis Simulation

A physics-based simulation environment is used to test the controller across multiple users and walking conditions.



Human ground reaction force and joint position, velocity, and torque data define the human portion of the model.

The prosthesis states and interaction forces are calculated together.

A prosthesis model with the interaction forces is forward simulated in 3D space with the controller.

Results

This work achieved human-like step placement behavior on a knee-ankle prosthesis in simulation for 22 subjects across 28 walking speeds and 35 speed transitions (1,386 walking conditions, 20,571 swing phase simulations).

These results demonstrate this control approach's potential to systematically generate user-specific controllers that enable stable, speed-adaptive prosthesis control, to support the daily activities of prosthesis users.

STEP PLACEMENT PREDICTION ERROR ACROSS SUBJECTS

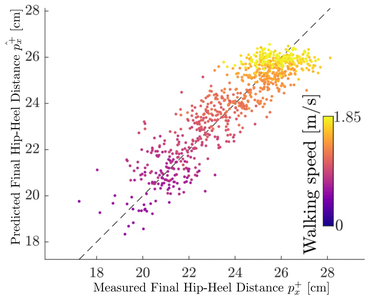
Prediction with Human Data Input $mean \pm s.d.$			
Step type	$\hat{p}_x^+$ [cm]	$\hat{p}_y^+$ [cm]	$\hat{T}_{SSP}$ [s]
All steps	1.9 ± 0.6	0.5 ± 0.1	0.032 ± 0.009
Steady-state	1.5 ± 0.5	0.4 ± 0.1	0.027 ± 0.008
Non-steady-state	4.3 ± 1.6	0.7 ± 0.2	0.066 ± 0.027
Prediction with Prosthesis Data Input $mean \pm s.d.$			
Step type	$\hat{p}_x^+$ [cm]	$\hat{p}_y^+$ [cm]	$\hat{T}_{SSP}$ [s]
All steps	1.9 ± 0.5	0.5 ± 0.1	0.033 ± 0.009
Steady-state	1.5 ± 0.5	0.4 ± 0.1	0.028 ± 0.008
Non-steady-state	4.3 ± 1.6	0.7 ± 0.2	0.066 ± 0.027

Step length, height, and duration are predicted similarly between inputs from the original human data and from the prosthesis in simulation.

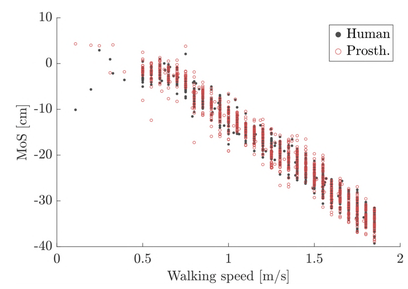
SWING TRAJECTORY TRACKING ERROR AT $\hat{T}_{SSP}$ ACROSS SUBJECTS

Tracking Error $mean \pm s.d.$			
Step type	p_x [cm]	p_y [cm]	θ_{foot} [deg]
All steps	0.4 ± 0.3	0.4 ± 0.1	0.9 ± 0.4
Steady-state	0.3 ± 0.1	0.4 ± 0.1	0.7 ± 0.2
Non-steady-state	1.0 ± 1.2	0.4 ± 0.2	1.6 ± 1.5

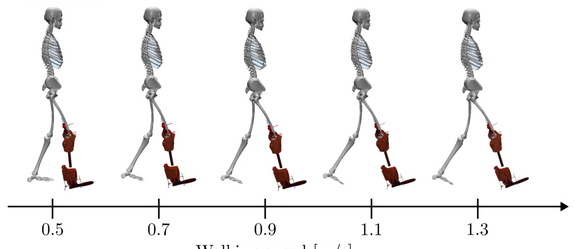
Tracking errors demonstrate the task-space controller successfully brings the prosthesis foot to its desired step length, height, and angle at heel-strike ($\hat{T}_{SSP}$).



Step length predictions ($\hat{p}_x^+$) align closely with actual step lengths (p_x^+), where the dashed line marks the ideal 1:1 relationship.



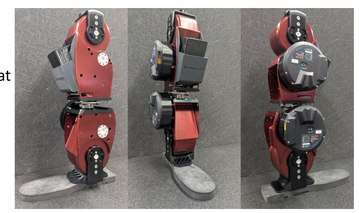
This controller achieved human-like gait stability patterns with respect to walking speed, as measured by Margin of Stability (MoS).



The controller realizes human-like step placement patterns with respect to walking speed.

Future work

- Expand prediction models to include perturbed walking.
- Develop a general prediction model that can be customized with minimal user-specific data.
- Incorporate a phase-variable to enable volitional control.
- Combine with a speed-adaptive stance controller and implement on hardware.



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